

Module 2-ANSI SQL Using MySQL

Create tables.sql

```
CREATE DATABASE EventPortalDB;

USE EventPortalDB;

-- Users Table
CREATE TABLE Users (
    user_id INT PRIMARY KEY AUTO_INCREMENT,
    full_name VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    city VARCHAR(100) NOT NULL,
    registration_date DATE NOT NULL
);

-- Events Table
CREATE TABLE Events (
    event_id INT PRIMARY KEY AUTO_INCREMENT,
    title VARCHAR(200) NOT NULL,
    description TEXT,
    city VARCHAR(100) NOT NULL,
    start_date DATETIME NOT NULL,
    end_date DATETIME NOT NULL,
    status ENUM('upcoming', 'completed', 'cancelled'),
    organizer_id INT,
    FOREIGN KEY (organizer_id) REFERENCES Users(user_id)
);

-- Sessions Table
CREATE TABLE Sessions (
    session_id INT PRIMARY KEY AUTO_INCREMENT,
    event_id INT,
    title VARCHAR(200) NOT NULL,
    speaker_name VARCHAR(100) NOT NULL,
    start_time DATETIME NOT NULL,
    end_time DATETIME NOT NULL,
    FOREIGN KEY (event_id) REFERENCES Events(event_id)
);

-- Registrations Table
CREATE TABLE Registrations (
    registration_id INT PRIMARY KEY AUTO_INCREMENT,
    user_id INT,
    event_id INT,
    registration_date DATE NOT NULL,
    FOREIGN KEY (user_id) REFERENCES Users(user_id),
    FOREIGN KEY (event_id) REFERENCES Events(event_id)
);

-- Feedback Table
CREATE TABLE Feedback (
    feedback_id INT PRIMARY KEY AUTO_INCREMENT,
    user_id INT,
    event_id INT,
    rating INT CHECK (rating BETWEEN 1 AND 5),
```

```

        comments TEXT,
        feedback_date DATE NOT NULL,
        FOREIGN KEY (user_id) REFERENCES Users(user_id),
        FOREIGN KEY (event_id) REFERENCES Events(event_id)
    );

-- Resources Table
CREATE TABLE Resources (
    resource_id INT PRIMARY KEY AUTO_INCREMENT,
    event_id INT,
    resource_type ENUM('pdf','image','link'),
    resource_url VARCHAR(255) NOT NULL,
    uploaded_at DATETIME NOT NULL,
    FOREIGN KEY (event_id) REFERENCES Events(event_id)
);

```

Insert tables.sql

```

USE EventPortalDB;

-- Users
INSERT INTO Users
(full_name,email,city,registration_date)
VALUES
('Alice Johnson','alice@example.com','New York','2024-12-01'),
('Bob Smith','bob@example.com','Los Angeles','2024-12-05'),
('Charlie Lee','charlie@example.com','Chicago','2024-12-10'),
('Diana King','diana@example.com','New York','2025-01-15'),
('Ethan Hunt','ethan@example.com','Los Angeles','2025-02-01');

-- Events
INSERT INTO Events
(title,description,city,start_date,end_date,status,organizer_id)
VALUES
(
    'Tech Innovators Meetup',
    'A meetup for tech enthusiasts.',
    'New York',
    '2025-06-10 10:00:00',
    '2025-06-10 16:00:00',
    'upcoming',
    1
),
(
    'AI & ML Conference',
    'Conference on AI and ML advancements.',
    'Chicago',
    '2025-05-15 09:00:00',
    '2025-05-15 17:00:00',
    'completed',
    3
),
(
    'Frontend Development Bootcamp',
    'Hands-on training on frontend tech.',
    'Los Angeles',

```

```

'2025-07-01 10:00:00',
'2025-07-03 16:00:00',
'upcoming',
2
);

-- Sessions
INSERT INTO Sessions
(event_id,title,speaker_name,start_time,end_time)
VALUES
(
1,
'Opening Keynote',
'Dr. Tech',
'2025-06-10 10:00:00',
'2025-06-10 11:00:00'
),
(
1,
'Future of Web Dev',
'Alice Johnson',
'2025-06-10 11:15:00',
'2025-06-10 12:30:00'
),
(
2,
'AI in Healthcare',
'Charlie Lee',
'2025-05-15 09:30:00',
'2025-05-15 11:00:00'
),
(
3,
'Intro to HTML5',
'Bob Smith',
'2025-07-01 10:00:00',
'2025-07-01 12:00:00'
);

-- Registrations
INSERT INTO Registrations
(user_id,event_id,registration_date)
VALUES
(1,1,'2025-05-01'),
(2,1,'2025-05-02'),
(3,2,'2025-04-30'),
(4,2,'2025-04-28'),
(5,3,'2025-06-15');

-- Feedback
INSERT INTO Feedback
(user_id,event_id,rating,comments,feedback_date)
VALUES
(
3,
2,
4,

```

```

'Great insights!',
'2025-05-16'
),
(
4,
2,
5,
'Very informative.',
'2025-05-16'
),
(
2,
1,
3,
'Could be better.',
'2025-06-11'
);

-- Resources
INSERT INTO Resources
(event_id, resource_type, resource_url, uploaded_at)
VALUES
(
1,
'pdf',
'https://portal.com/resources/tech_meetup_agenda.pdf',
'2025-05-01 10:00:00'
),
(
2,
'image',
'https://portal.com/resources/ai_poster.jpg',
'2025-04-20 09:00:00'
),
(
3,
'link',
'https://portal.com/resources/html5_docs',
'2025-06-25 15:00:00'
);

```

Exercises:

1. User Upcoming Events

Show a list of all upcoming events a user is registered for in their city, sorted by date.

Source Code

```

SELECT u.full_name,
       e.title,
       e.city,
       e.start_date
FROM Users u
JOIN Registrations r ON u.user_id = r.user_id
JOIN Events e ON r.event_id = e.event_id
WHERE e.status = 'upcoming'
      AND u.city = e.city
ORDER BY e.start_date;

```

create_tables insert_data SQL File 2* x

Limit to 1000 rows

```

1 • SELECT u.full_name,
2         e.title,
3         e.city,
4         e.start_date
5 FROM Users u
6 JOIN Registrations r ON u.user_id = r.user_id
7 JOIN Events e ON r.event_id = e.event_id
8 WHERE e.status = 'upcoming'
9        AND u.city = e.city
10 ORDER BY e.start_date;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	full_name	title	city	start_date
▶	Alice Johnson	Tech Innovators Meetup	New York	2025-06-10 10:00:00
	Ethan Hunt	Frontend Development Bootcamp	New York	2025-06-10 10:00:00

Frontend Development Bootcamp

2. Top Rated Events

Identify events with the highest average rating, considering only those that have received at least 10 feedback submissions.

Source Code

```

SELECT e.event_id,
       e.title,
       AVG(f.rating) AS avg_rating,
       COUNT(*) AS feedback_count
FROM Events e
JOIN Feedback f ON e.event_id = f.event_id
GROUP BY e.event_id, e.title
HAVING COUNT(*) >= 10
ORDER BY avg_rating DESC;

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	user_id	full_name	email	city	registration_date
▶	1	Alice Johnson	alice@example.com	New York	2024-12-01
	2	Bob Smith	bob@example.com	Los Angeles	2024-12-05
	3	Charlie Lee	charlie@example.com	Chicago	2024-12-10
	4	Diana King	diana@example.com	New York	2025-01-15
	5	Ethan Hunt	ethan@example.com	Los Angeles	2025-02-01
*	NULL	NULL	NULL	NULL	NULL

3. Inactive Users

Retrieve users who have not registered for any events in the last 90 days.

Source Code

```

SELECT *
FROM Users
WHERE user_id NOT IN (
    SELECT DISTINCT user_id

```

```
FROM Registrations
WHERE registration_date >= CURDATE() - INTERVAL 90 DAY
);
```

Result Grid					
Filter Rows:					
Edit:					
Export/Import:					
Wrap Cell Content:					
	user_id	full_name	email	city	registration_date
▶	1	Alice Johnson	alice@example.com	New York	2024-12-01
	2	Bob Smith	bob@example.com	Los Angeles	2024-12-05
	3	Charlie Lee	charlie@example.com	Chicago	2024-12-10
	4	Diana King	diana@example.com	New York	2025-01-15
	5	Ethan Hunt	ethan@example.com	Los Angeles	2025-02-01
*	NULL	NULL	NULL	NULL	NULL

4. Peak Session Hours

Count how many sessions are scheduled between 10 AM to 12 PM for each event.

Source Code

```
SELECT e.title,
       COUNT(s.session_id) AS session_count
FROM Events e
LEFT JOIN Sessions s
ON e.event_id = s.event_id
WHERE TIME(s.start_time) BETWEEN '10:00:00' AND '12:00:00'
GROUP BY e.event_id, e.title;
```

Result Grid		
Filter Rows:		
Export:		
	title	session_count
▶	Tech Innovators Meetup	2
	Frontend Development Bootcamp	1

5. Most Active Cities

List the top 5 cities with the highest number of distinct user registrations.

Source Code

```
SELECT u.city,
       COUNT(DISTINCT r.user_id) AS registrations
FROM Users u
JOIN Registrations r
ON u.user_id = r.user_id
GROUP BY u.city
ORDER BY registrations DESC
LIMIT 5;
```

Result Grid			Filter Rows:
	city	registrations	
▶	Los Angeles	2	
	New York	2	
	Chicago	1	

6. Event Resource Summary

Generate a report showing the number of resources (PDFs, images, links) uploaded for each event.

Source Code

```
SELECT e.title,
       COUNT(CASE WHEN resource_type='pdf' THEN 1 END) AS pdf_count,
       COUNT(CASE WHEN resource_type='image' THEN 1 END) AS image_count,
       COUNT(CASE WHEN resource_type='link' THEN 1 END) AS link_count
FROM Events e
LEFT JOIN Resources r
ON e.event_id = r.event_id
GROUP BY e.event_id, e.title;
```

Result Grid					Filter Rows:	Export:	Wrap Cell Content:
	title	pdf_count	image_count	link_count			
▶	Tech Innovators Meetup	1	0	0			
	AI & ML Conference	0	1	0			
	Frontend Development Bootcamp	0	0	1			

7. Low Feedback Alerts

List all users who gave feedback with a rating less than 3, along with their comments and associated event names.

Source Code

```
SELECT u.full_name,
       e.title,
       f.rating,
       f.comments
FROM Feedback f
JOIN Users u ON f.user_id=u.user_id
JOIN Events e ON f.event_id=e.event_id
WHERE f.rating < 3;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	title	total_sessions			
▶	Tech Innovators Meetup	2			
	Frontend Development Bootcamp	1			

8. Sessions per Upcoming Event

Display all upcoming events with the count of sessions scheduled for them.

Source Code

```
SELECT e.title,
       COUNT(s.session_id) AS total_sessions
FROM Events e
LEFT JOIN Sessions s
ON e.event_id=s.event_id
WHERE e.status='upcoming'
GROUP BY e.event_id,e.title;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	title	total_sessions			
▶	Tech Innovators Meetup	2			
	Frontend Development Bootcamp	1			

9. Organizer Event Summary

For each event organizer, show the number of events created and their current status (upcoming, completed, cancelled).

Source Code

```
SELECT u.full_name,
       e.status,
       COUNT(*) AS total_events
FROM Events e
JOIN Users u
ON e.organizer_id=u.user_id
GROUP BY u.full_name,e.status;
```

Result Grid				Filter Rows:	Export:	W
	full_name	status	total_events			
▶	Alice Johnson	Alice Johnson				
	Charlie Lee	completed	1			
	Bob Smith	upcoming	1			

10. Feedback Gap

Identify events that had registrations but received no feedback at all.

Source Code

```
SELECT e.title
FROM Events e
```



```

JOIN Registrations r
ON e.event_id=r.event_id
LEFT JOIN Feedback f
ON e.event_id=f.event_id
WHERE f.feedback_id IS NULL
GROUP BY e.event_id,e.title;

```

Result Grid		Filter Rows:
	title	
▶	Frontend Development Bootcamp	

11. Daily New User Count

Find the number of users who registered each day in the last 7 days.

Source Code

```

SELECT registration_date,
       COUNT(*) AS new_users
FROM Users
WHERE registration_date >= CURDATE()-INTERVAL 7 DAY
GROUP BY registration_date;

```

Result Grid		Filter Rows:
	title	
▶	Frontend Development Bootcamp	

12. Event with Maximum Sessions

List the event(s) with the highest number of sessions.

Source Code

```

SELECT e.title,
       COUNT(s.session_id) AS total_sessions
FROM Events e
JOIN Sessions s
ON e.event_id=s.event_id
GROUP BY e.event_id,e.title
HAVING COUNT(s.session_id)=(
    SELECT MAX(session_count)
    FROM (
        SELECT COUNT(*) AS session_count
        FROM Sessions
        GROUP BY event_id
    ) x
);

```

Result Grid			Filter Rows:
	title	total_sessions	
▶	Tech Innovators Meetup	2	2

13. Average Rating per City

Calculate the average feedback rating of events conducted in each city.

Source Code

```

SELECT e.city,
       AVG(f.rating) AS avg_rating
FROM Events e
JOIN Feedback f
ON e.event_id=f.event_id
GROUP BY e.city;

```

Result Grid		
	city	avg_rating
▶	New York	3.0000
	Chicago	4.5000

14. Most Registered Events

List top 3 events based on the total number of user registrations.

Source Code

```
SELECT e.title,
       COUNT(r.registration_id) AS registrations
FROM Events e
JOIN Registrations r
ON e.event_id=r.event_id
GROUP BY e.event_id,e.title
ORDER BY registrations DESC
LIMIT 3;
```

Result Grid		
	title	registrations
▶	Tech Innovators Meetup	2
	AI & ML Conference	2
	Frontend Development Bootcamp	1

15. Event Session Time Conflict

Identify overlapping sessions within the same event (i.e., session start and end times that conflict).

Source Code

```
SELECT s1.event_id,
       s1.title,
       s2.title AS conflicting_session
FROM Sessions s1
JOIN Sessions s2
ON s1.event_id=s2.event_id
AND s1.session_id < s2.session_id
AND s1.start_time < s2.end_time
AND s1.end_time > s2.start_time;
```

Result Grid		
	title	registrations
▶	Tech Innovators Meetup	2
	AI & ML Conference	2
	Frontend Development Bootcamp	1

16. Unregistered Active Users

Find users who created an account in the last 30 days but haven't registered for any events.

Source Code

```
SELECT *
FROM Users u
WHERE registration_date >= CURDATE()-INTERVAL 30 DAY
AND NOT EXISTS (
  SELECT 1
```

```
FROM Registrations r
WHERE r.user_id=u.user_id
);
```

	user_id	full_name	email	city	registration_date
*	NULL	NULL	NULL	NULL	NULL

17. Multi-Session Speakers

Identify speakers who are handling more than one session across all events.

Source Code

```
SELECT speaker_name,
       COUNT(*) AS sessions_handled
FROM Sessions
GROUP BY speaker_name
HAVING COUNT(*) > 1;
```

	user_id	full_name	email	city	registration_date
*	NULL	NULL	NULL	NULL	NULL

18. Resource Availability Check

List all events that do not have any resources uploaded.

Source Code

```
SELECT e.title
FROM Events e
LEFT JOIN Resources r
ON e.event_id=r.event_id
WHERE r.resource_id IS NULL;
```

	user_id	full_name	email	city	registration_date
*	NULL	NULL	NULL	NULL	NULL

19. Completed Events with Feedback Summary

For completed events, show total registrations and average feedback rating.

Source Code

```
SELECT e.title,
       COUNT(DISTINCT r.registration_id) AS registrations,
       AVG(f.rating) AS avg_rating
FROM Events e
LEFT JOIN Registrations r
ON e.event_id=r.event_id
LEFT JOIN Feedback f
ON e.event_id=f.event_id
WHERE e.status='completed'
GROUP BY e.event_id,e.title;
```

	title	registrations	avg_rating
▶	AI & ML Conference	2	4.5000

20. User Engagement Index

For each user, calculate how many events they attended and how many feedbacks they submitted.

Source Code

```

SELECT u.full_name,
       COUNT(DISTINCT r.event_id) AS attended_events,
       COUNT(DISTINCT f.feedback_id) AS feedbacks_submitted
FROM Users u
LEFT JOIN Registrations r
ON u.user_id=r.user_id
LEFT JOIN Feedback f
ON u.user_id=f.user_id
GROUP BY u.user_id,u.full_name;

```

Result Grid			
Filter Rows:			
Export:			
	full_name	attended_events	feedbacks_submitted
▶	Alice Johnson	1	0
	Bob Smith	1	1
	Charlie Lee	1	1
	Diana King	1	1
	Ethan Hunt	1	0

21. Top Feedback Providers

List top 5 users who have submitted the most feedback entries.

Source Code

```

SELECT u.full_name,
       COUNT(*) AS feedback_count
FROM Users u
JOIN Feedback f
ON u.user_id=f.user_id
GROUP BY u.user_id,u.full_name
ORDER BY feedback_count DESC
LIMIT 5;

```

Result Grid		
Filter Rows:		
	full_name	feedback_count
▶	Bob Smith	1
	Charlie Lee	1
	Diana King	1

22. Duplicate Registrations Check

Detect if a user has been registered more than once for the same event.

Source Code

```

SELECT user_id,
       event_id,
       COUNT(*) AS duplicate_count
FROM Registrations
GROUP BY user_id,event_id
HAVING COUNT(*) > 1;

```

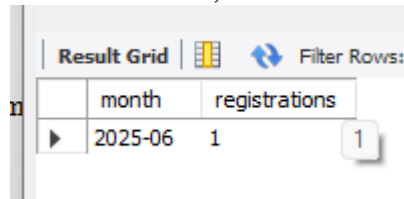
Result Grid		
Filter Rows:		
	full_name	feedback_count
▶	Bob Smith	1
	Charlie Lee	1
	Diana King	1

23. Registration Trends

Show a month-wise registration count trend over the past 12 months.

Source Code

```
SELECT DATE_FORMAT(registration_date,'%Y-%m') AS month,
       COUNT(*) AS registrations
FROM Registrations
WHERE registration_date >= CURDATE()-INTERVAL 12 MONTH
GROUP BY month
ORDER BY month;
```



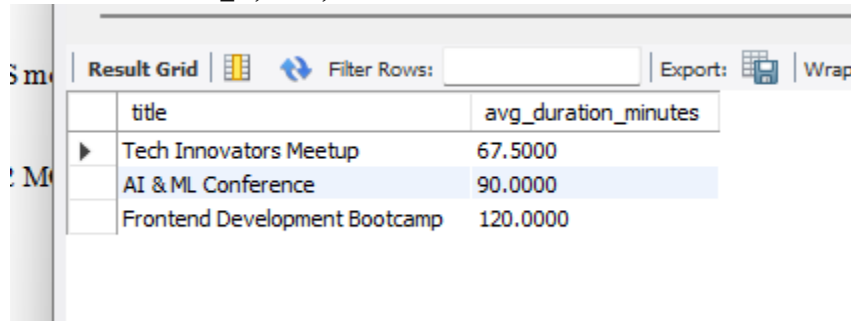
month	registrations
2025-06	1

24. Average Session Duration per Event

Compute the average duration (in minutes) of sessions in each event.

Source Code

```
SELECT e.title,
       AVG(
         TIMESTAMPDIFF(
           MINUTE,
           s.start_time,
           s.end_time
         )
       ) AS avg_duration_minutes
FROM Events e
JOIN Sessions s
ON e.event_id=s.event_id
GROUP BY e.event_id,e.title;
```



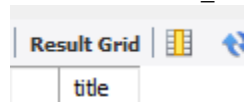
title	avg_duration_minutes
Tech Innovators Meetup	67.5000
AI & ML Conference	90.0000
Frontend Development Bootcamp	120.0000

25. Events Without Sessions

List all events that currently have no sessions scheduled under them.

Source Code

```
SELECT e.title
FROM Events e
LEFT JOIN Sessions s
ON e.event_id=s.event_id
WHERE s.session_id IS NULL;
```



title
